Roostoo ML Intern Assignment

Atharva Moghe · Predictive ML Trading Assistant assessment · 28 October 2026
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Goal. Help a new BTC/USDT trader take **fewer and better trades**. Every 4 hours the agent answers four questions: *what is the market like, is a trade worth taking after my own fees, how big should it be and where do I exit, and should I trust the agent right now?*

1. About the model

1.1 Prediction target and horizon

- **Target.** For a long entry now, **calibrated probabilities of three outcomes in the next 24 hours**: *take-profit first* ($+1.5\sigma$), *stop first* (-1σ), or *neither*. σ is the current typical 24h move, so the barriers scale with volatility.
- **Why it helps.** A beginner needs a **decision, not a forecast**. With the trader's own fees the probabilities give the **expected value of this trade for this trader**, $\text{EV} = (1.5\,p_{\text{TP}} - p_{\text{SL}} + \bar{r}_{\text{TO}}\,p_{\text{TO}})\,\sigma - \text{fees}$. Its sign decides *take or skip*; the same σ sets the stop, target and position size.
- **Horizon: 24h on 1h bars**, picked from a grid *fixed in advance* (15m/1h/4h bars × 4h/12h/24h). 24h was best at every bar size and 4h worst (Figure 1a), exactly what fees predict.

1.2 Data and features

- **Data.** Free **Binance** data, Jan 2021 – Oct 2026: BTC and ETH spot (15 min), plus *futures funding rates, open interest and long/short ratios.*
- **Features. 39 features in six groups**, each a hypothesis. **Volatility** was strongest; *order flow*, *futures positioning* (crowded leveraged longs) and *ETH vs BTC* added value; short-term *momentum* never did.
- **No leakage.** Features use only data available at that moment; feature selection is fitted on training folds only.

1.3 Model and baseline

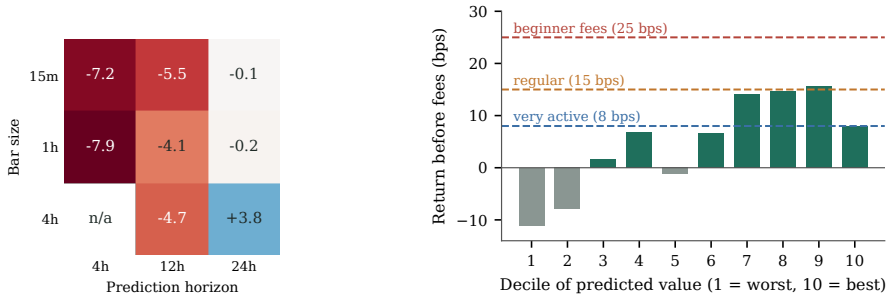
- **Model.** A **shallow three-class LightGBM** (depth 3), with weights for overlapping labels and **isotonic calibration**, because the decision compares a probability with a fee.
- **Why this model.** The inputs are *tabular, mixed-scale and partly missing*, and the useful signals are *interactions* (high funding *and* rising open interest means a crowded market). **Gradient-boosted trees** handle all of this natively and retrain in seconds. They are kept **shallow** because there are only about **2,000 independent 24h outcomes**, where deep trees or neural networks would fit noise. *Three classes* let EV account for stop-outs and trades that go nowhere.
- **Starting model.** *Logistic regression* on the same features. LightGBM beat it only at 24h and only slightly on the final test (-15.8% vs -17.9%), so logistic regression stays as the **fallback**.
- **Baselines.** Buy-and-hold, every trade, majority-class probabilities, a **random filter with the same number of trades**, **shuffled-label retrains**, and two published strategies (4-week trend, funding carry).

1.4 Validation and backtest design

- **Walk-forward**: train on everything up to a date, test on the next 6 months, **8 periods** from mid-2022; overlapping labels are *purged*; the **last 6 months held out**.
- **Realistic costs**: **0.25% round trip** on spot; futures limit orders fill only if price trades through; funding charged.
- **Checks**: prediction and trading metrics reported separately, with *bootstrap ranges*, a *shuffled-label null* and a *fee sweep.*

Results. Direction is **weak** (AUC ≈ 0.55), but the **ranking of trades is real**: it beat 97% of shuffled retrains and held out of sample (Figure 1b). The average edge (**0.09%**) is **below beginner fees (0.25%)**; breakeven is about 0.10%. The best configuration **lost 29% on the final 6 months**: the edge was strong in 2023–2024 and faded.

The model's **real strength is predicting the size of the next move**. A next-24h **volatility forecast beats every standard baseline, including HAR-RV** (Table 1, Figure 2); trade-count surges, open-interest changes and ETH vs BTC are top features that a price-only model cannot see. The coach uses it for *stops, sizing and "wild day" warnings.*



(a) Net return per trade (bps). **(b)** Pre-fee return by decile vs fee levels.

Figure 1: (a) Only the 24h horizon is near or above zero. (b) The model ranks trades (top deciles +14–16 bps before fees), but only a very-low-fee trader clears the bar.

Table 1: Next-24h volatility forecasts (QLIKE loss, lower is better).

Model	QLIKE, 8 periods	vs HAR	QLIKE, final 6m	R^2 (log), final 6m
Yesterday's volatility	0.493	−31%	0.374	−0.37
EWMA (3-day)	0.437	−16%	0.304	−0.05
HAR-RV (benchmark)	0.376	–	0.274	0.01
LightGBM, our features	**0.349**	**+7%** (t=1.6)	**0.221**	**0.35**

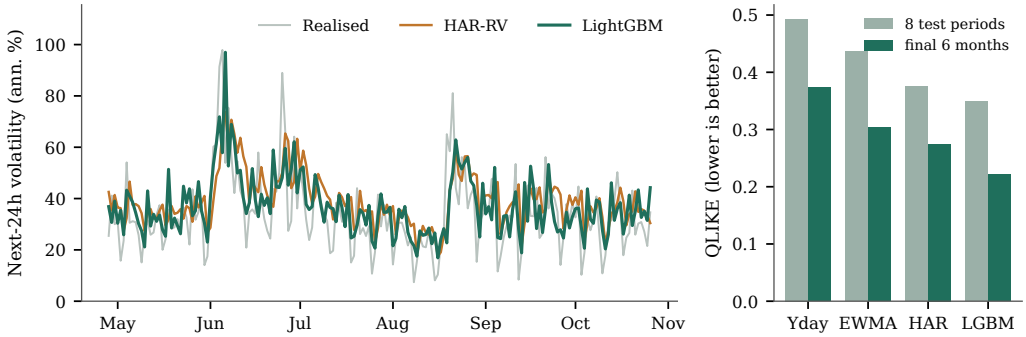


Figure 2: Realised vs forecast volatility (final 6 months) and forecast error by model. Gain over HAR: borderline over 8 periods, significant on the final 6 months (+19%, t=2.3).

1.5 How the agent works and what the trader sees

The output must be **simple**, since the user is new to crypto. Figure 3 shows the loop the agent runs every 4 hours.

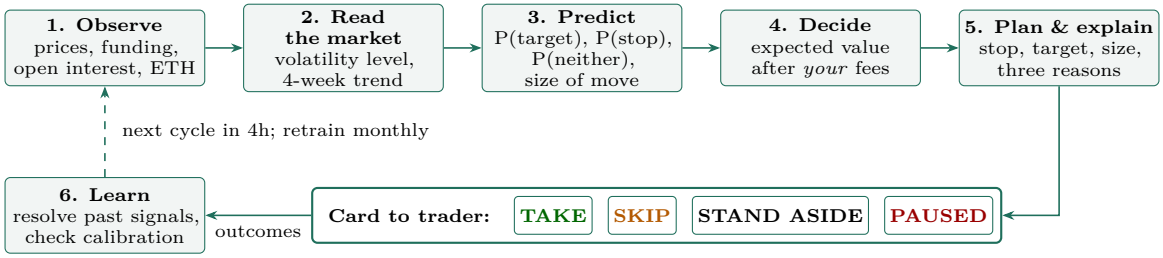


Figure 3: The agent's 4-hour decision loop. Steps 1–5 turn the latest data into one card for the trader. Step 6 compares past predictions with what actually happened: if they have drifted, the next cards are PAUSED until accuracy recovers; the model is also retrained every month on newly resolved data.

Each card is in one of four states: **TAKE** (edge beats *your* fees), **SKIP** (edge exists but fees remove it; the card shows the fee level that would make it worthwhile), **STAND ASIDE** (no edge; hold in an uptrend, cash in a downtrend) or **PAUSED** (recent predictions drifted from outcomes). A real card from the replay, for a regular trader:

> **TAKE — leaning up for the next 24h.** Confidence: high (top 10% of setups).
> Odds: target first 12%, stop first 11%, neither 76%. Expected +0.27%, your fees 0.15%, **net +0.12%**.
> Buy \$84,736 · take profit \$86,498 · stop \$83,581 · close after 24h · size 73% of account (the stop costs 1%).
> Why: funding is low this month; ETH is lagging BTC; weekly volatility is low.

The trader picks a **profile**, which changes which signals are shown and how they are sized:

	Beginner	Regular	Very active
Signals shown	Top 10% confidence, clearly above fees	Medium and high confidence	Any edge after fees
Risk per trade	0.5%, at most half the account	1%	1%
Typical fees	0.25% spot	0.15%	0.08% futures
Falling markets	Cash; explains why not to short	Explains shorts only	Short view, no leverage
Final 6 months, coach	$\mathbf{-9.1}\%$ (20 trades)	$\mathbf{-3.5}\%$ (45 trades)	$\mathbf{-1.8}\%$ (68 trades)
Final 6 months, every signal	-61%	-39%	-24%

Check my idea grades the trader's own plan from A to D (*chasing a pump, a stop inside normal noise, too much at risk*), then shows what actually happened. A **weekly report** and a **track record** on the price chart (losers included) show whether the agent's confidence matched outcomes.

1.6 Salient features of the model

- **Decides, does not forecast**: a trade decision with entry, stop, target and size, not an up/down call.
- **Personal**: the same moment can be TAKE at 0.08% fees and SKIP at 0.25%; profiles set risk and signal strictness.
- **Built on what the data supports**: move size, which *beats HAR-RV*, drives stops, sizing and warnings.
- **Calibrated and self-monitoring**: it *pauses itself* when predictions drift and resumes when they recover.
- **Honestly benchmarked**: purged walk-forward, real fees and fills, random-filter and shuffled-label tests.
- **Explains itself**: three plain-language reasons per card and a public track record that includes losses.

1.7 Risks, limitations and next steps

- **Edge decay**: positive in 2023–2024, negative since; handled by *monthly retraining* and the *automatic pause*.
- **Fees above the edge** (0.09% vs 0.25%): decisions use the trader's own fees, so beginners mostly *skip*.
- **Overfitting the search** (19 runs): grid fixed in advance, every run logged, random and shuffled benchmarks.
- **Reused test period**: flagged; *live paper trading* is the remaining clean test.
- **Long-only blind spot**: shorts made money only in falling periods and lost 93% overall, so only very active traders see them.

Limitations. BTC only; confidence levels were not informative in the final 6 months. **With more time:** paper-trade the agent live and build a direction model on top of the volatility forecast.

AI assistance. Claude (Anthropic) was used to help write parts of the codebase and to draft and edit this document. The research direction, design decisions and final judgement on every result are mine, and I take full responsibility for the content.